Hyperbolic Image-Text Representations

자연어 처리팀:
조해창(발표자), 박산희, 황예진, 황지현, 정강민

Content

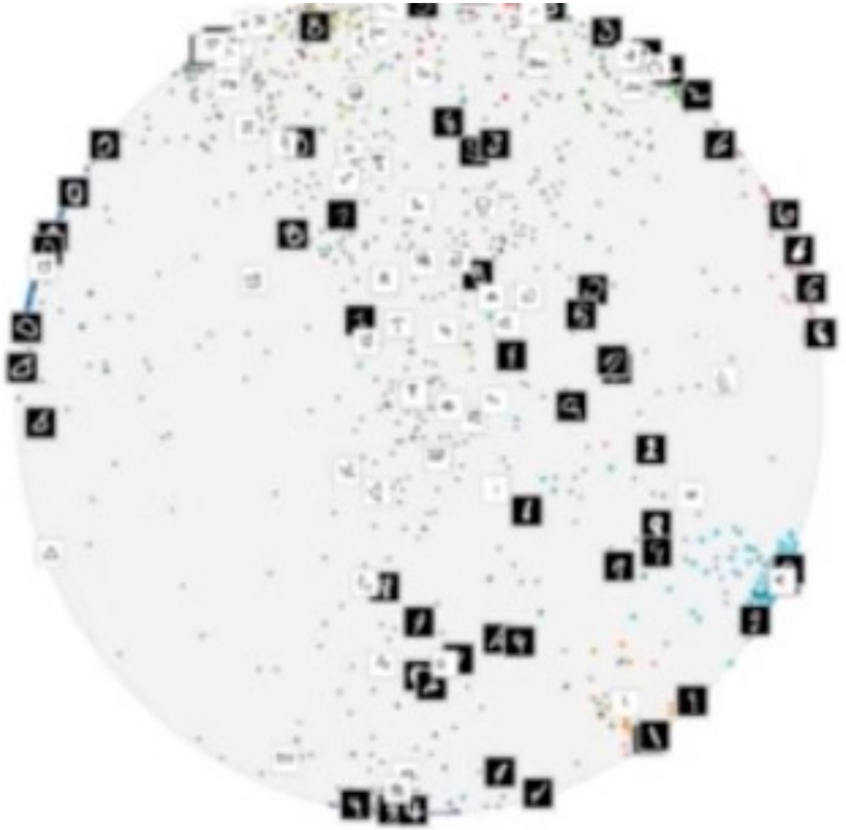
- Motivation
- Preliminaries
- Experiments
- Conclusion

Image traversals with YFCC captions.

(1)

MERU	CLIP
leopard and stig have a beautiful piano at their home.	*loki is a 1 year old bengal cat.*
merlin wasn't impressed to leave the last house and his precious cat grass	↓
my parents cat 'barry' loves being photographed!	↓
house cat posing	↓
mr . bo-majed	↓
our cat, our love our third member of our family. :)	↓
why are you taking pictures? it's dilo don't fill it up. :)	↓
[ROOT]	[ROOT]


나무
집
개
눈

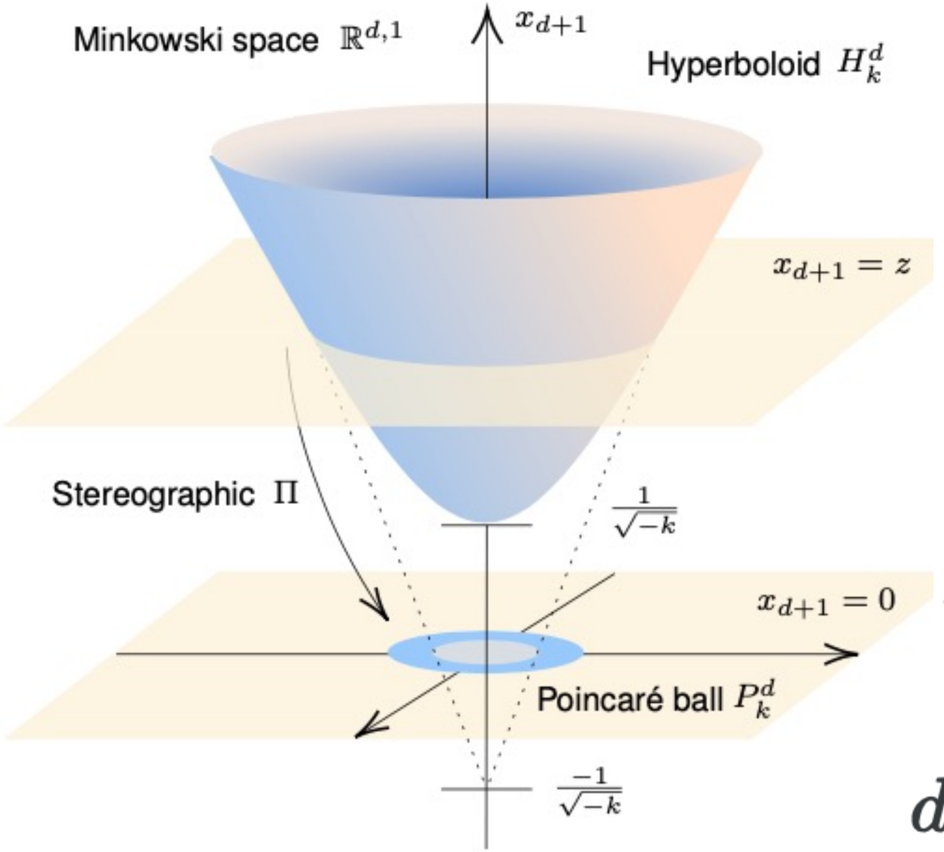


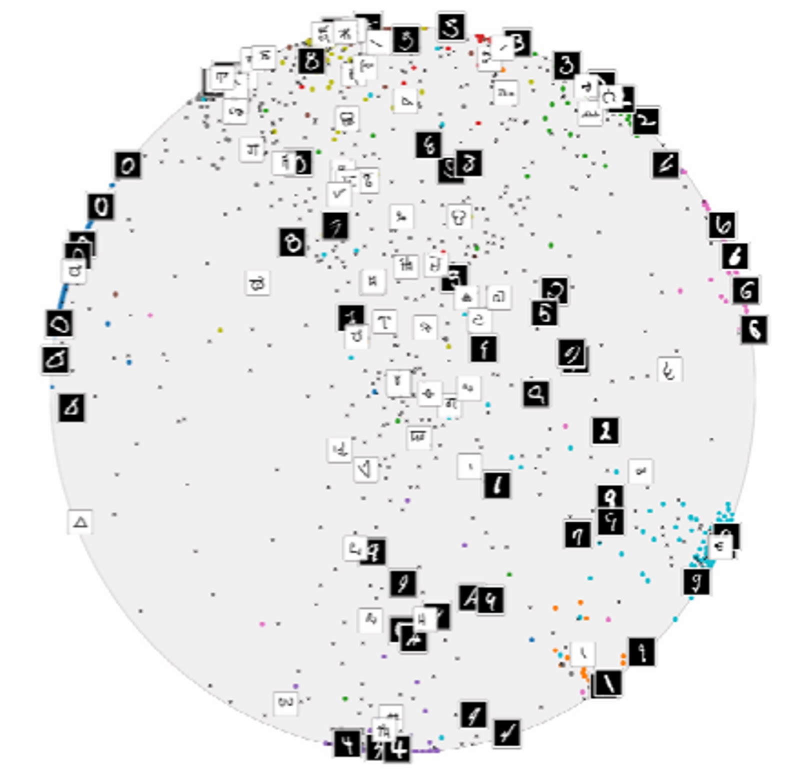
Hyperbolic Image Embeddings

CVPR 2020

이미지처리팀

류채은, 강인하, 김병현, 김선옥, 김준철, 김현진, 남종우, 안종식, 이찬혁, 이희재, 조경민, 최승준, 현청천



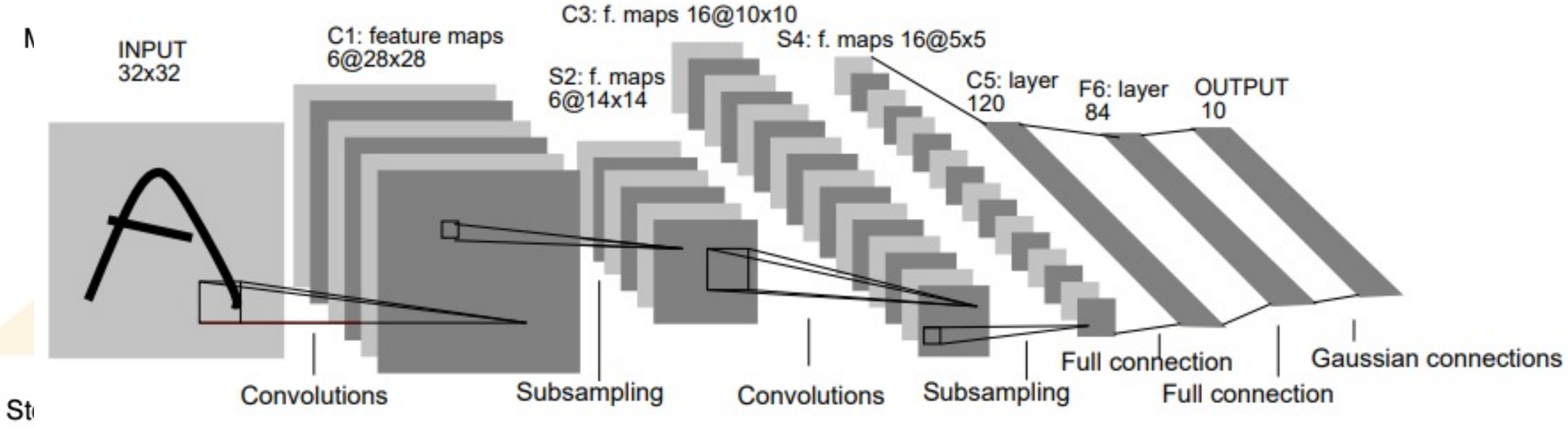


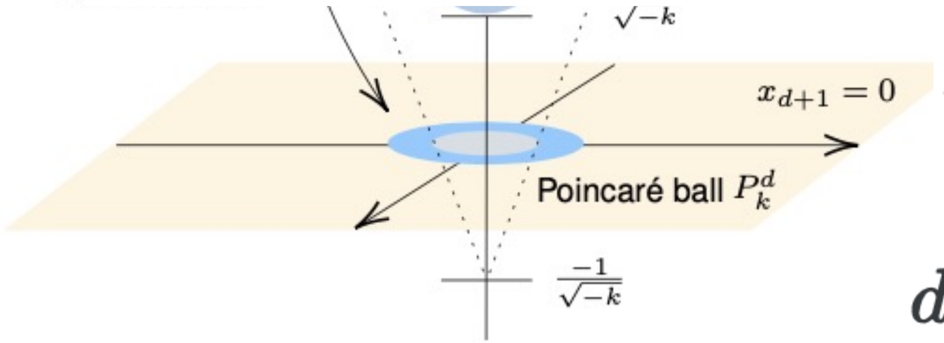
$$r = \sqrt{x^2 + y^2}$$

$$ds^2 = \frac{dx^2 + dy^2}{(1 - r^2)^2}$$

$$d(\mathbf{u}, \mathbf{v}) = \cosh^{-1}\left(1 + 2\frac{\|\mathbf{u} - \mathbf{v}\|^2}{(1 - \|\mathbf{u}\|^2)(1 - \|\mathbf{v}\|^2)}\right)$$

$$d_{\mathbb{D}}(\mathbf{x}, \mathbf{y}) = \operatorname{arccosh}\left(1 + 2\frac{\|\mathbf{x} - \mathbf{y}\|^2}{(1 - \|\mathbf{x}\|^2)(1 - \|\mathbf{y}\|^2)}\right).$$





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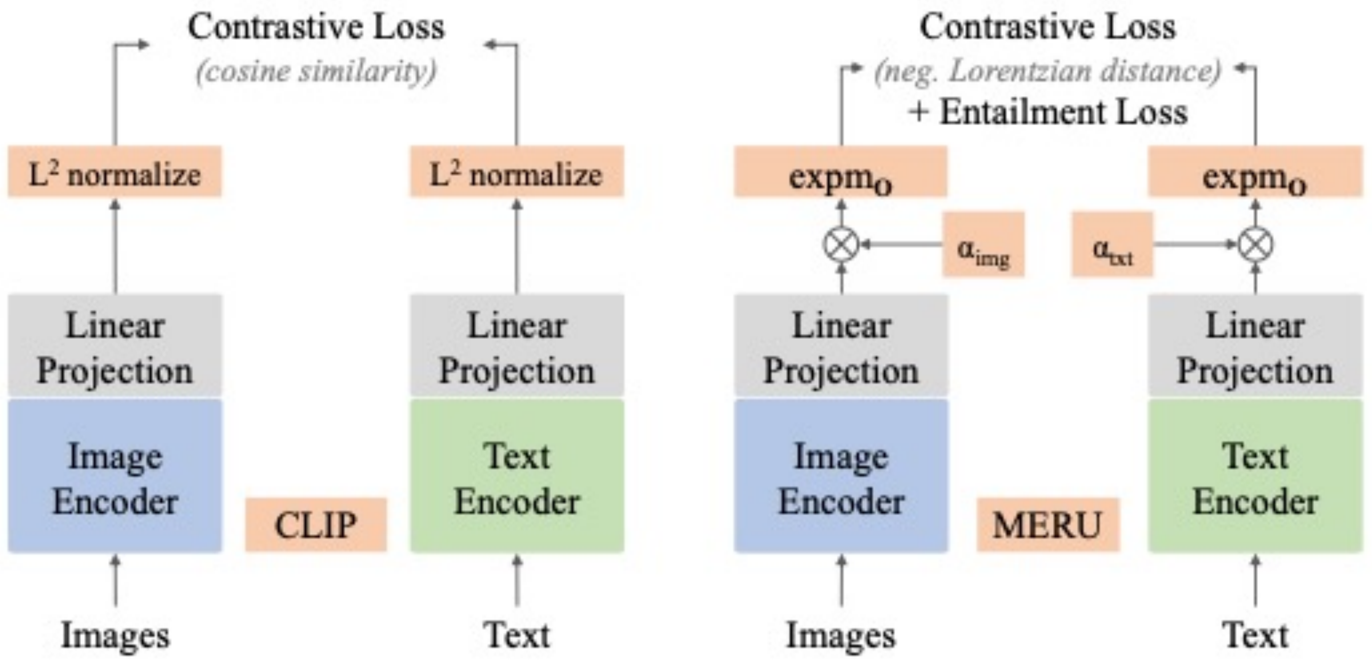


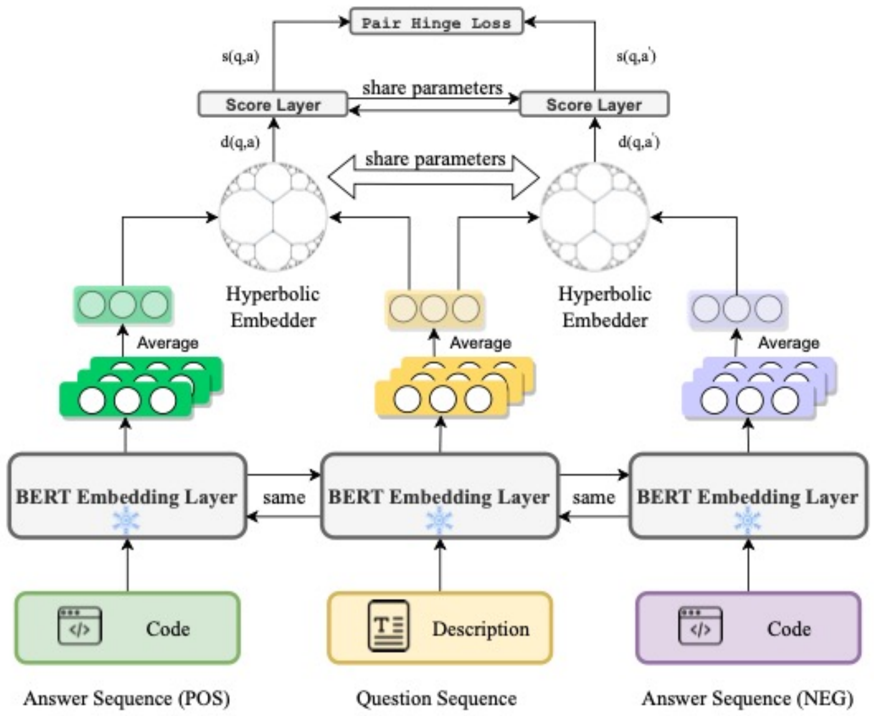
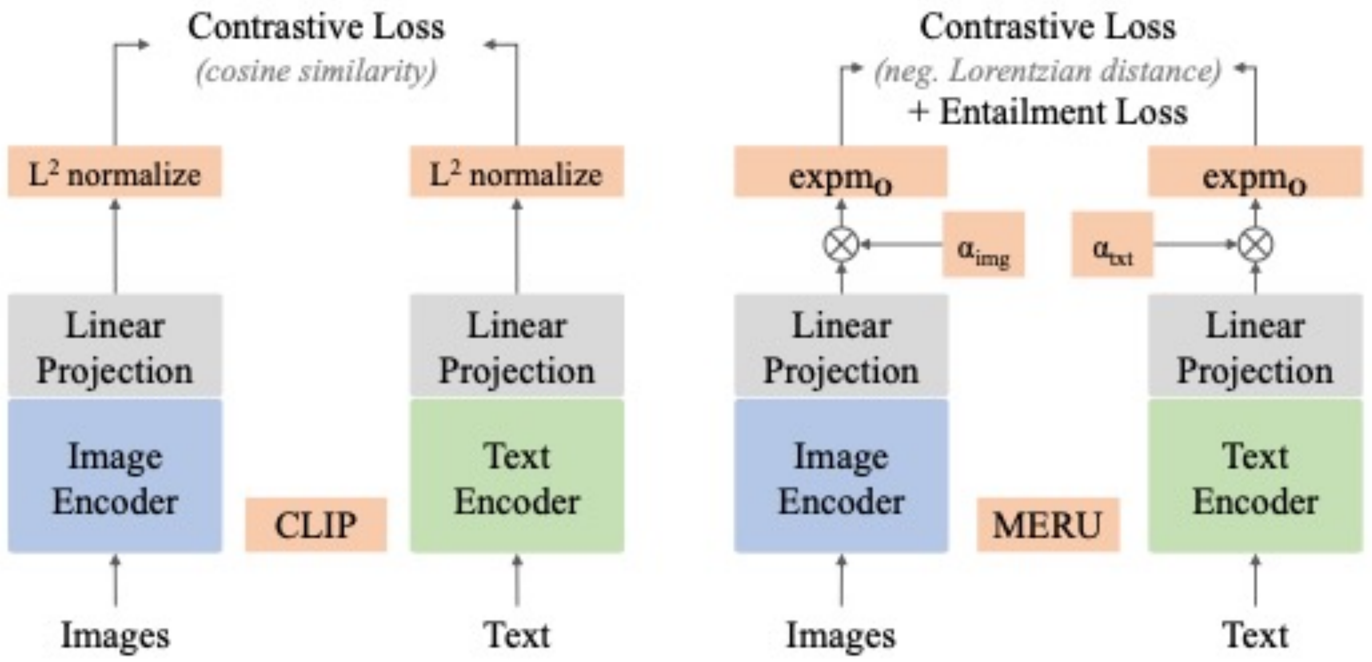


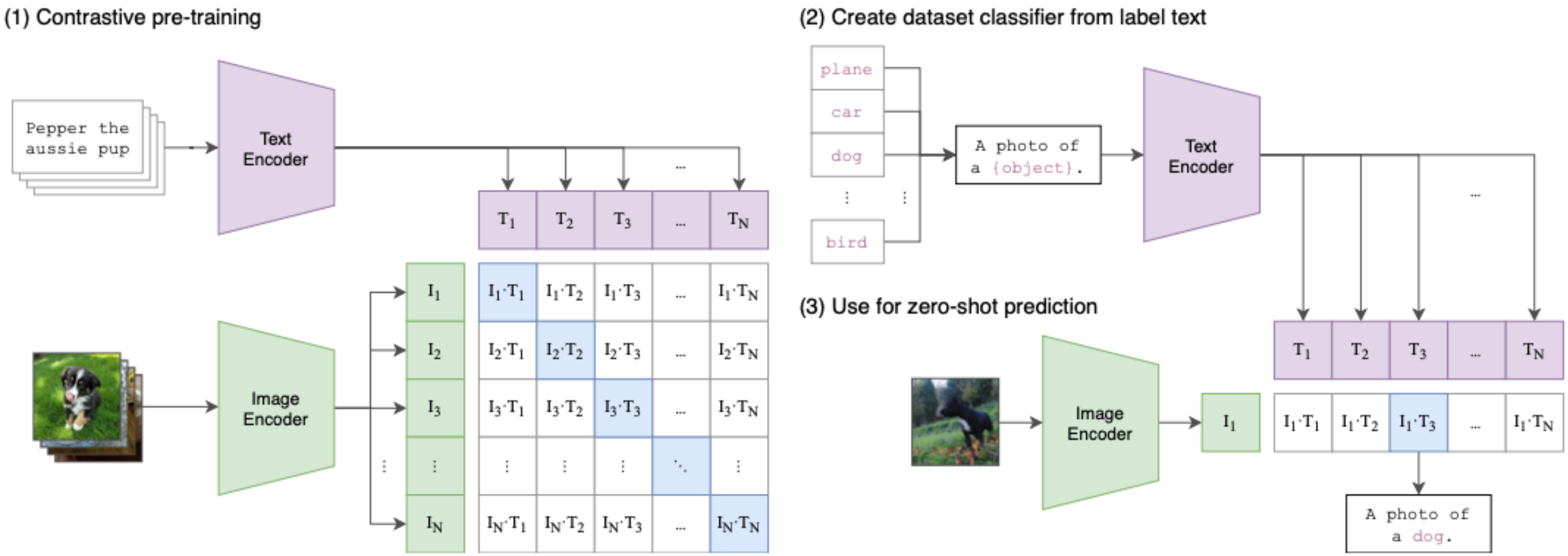
Figure 1: Architecture of `HyCoQA`.

Hyperbolic Image-Text Representations

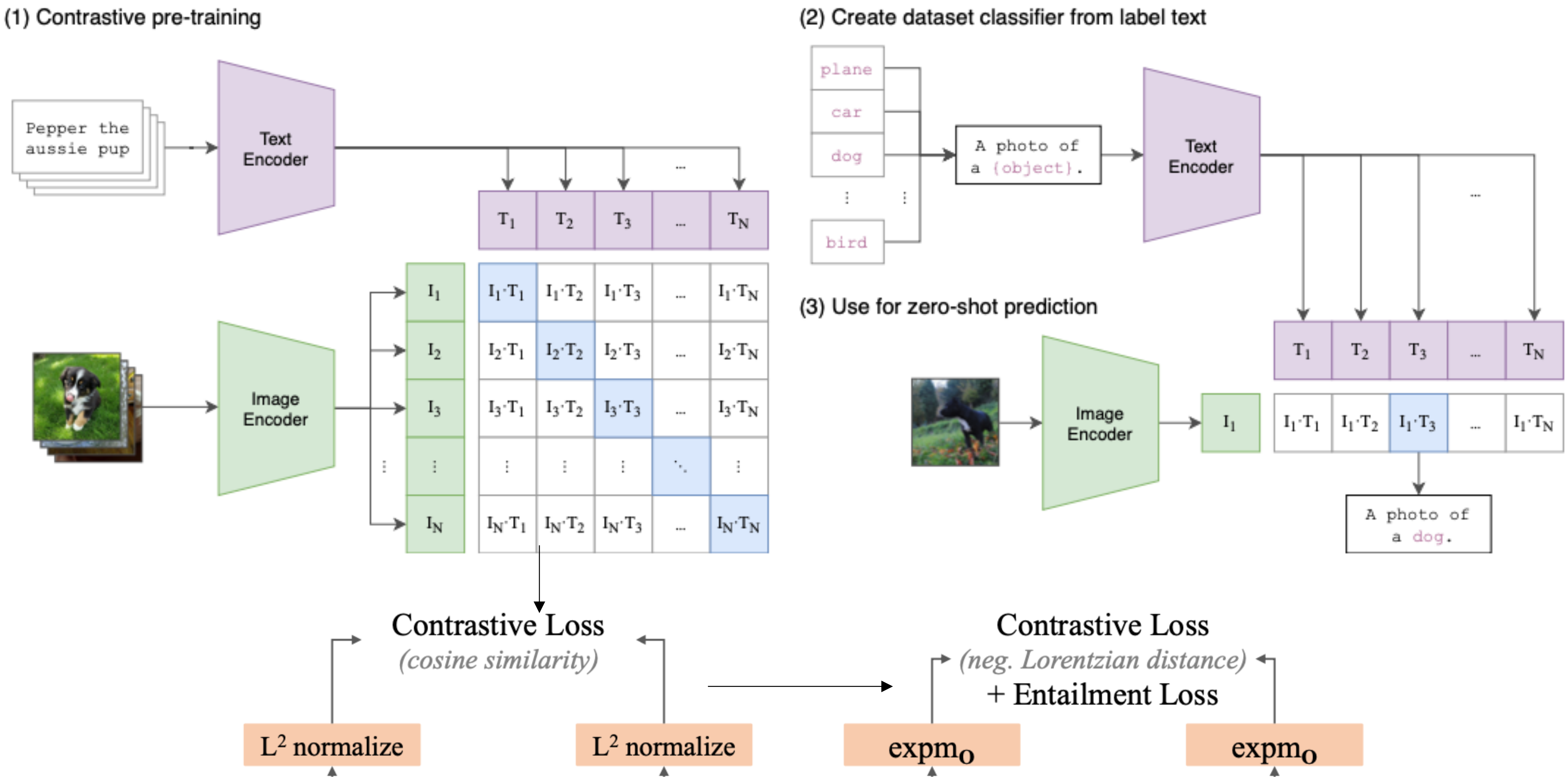


백문이 불여일견(百聞而 不如一見)이요,

'an image is worth a thousand words'



(1) Contrastive pre-training
Pepper the aussie pup
Text Encoder
Image Encoder
T_1 T_2 T_3 ... T_N
I_1 I_2 I_3 ... I_N
$I_1 \cdot T_1$ $I_1 \cdot T_2$ $I_1 \cdot T_3$ ... $I_1 \cdot T_N$
$I_2 \cdot T_1$ $I_2 \cdot T_2$ $I_2 \cdot T_3$ ... $I_2 \cdot T_N$
$I_3 \cdot T_1$ $I_3 \cdot T_2$ $I_3 \cdot T_3$ ... $I_3 \cdot T_N$
$I_N \cdot T_1$ $I_N \cdot T_2$ $I_N \cdot T_3$ ... $I_N \cdot T_N$
(2) Create dataset classifier from label text
plane
car
dog
bird
A photo of a {object}.
Text Encoder
(3) Use for zero-shot prediction
Image Encoder
I_1
T_1 T_2 T_3 ... T_N
$I_1 \cdot T_1$ $I_1 \cdot T_2$ $I_1 \cdot T_3$ ... $I_1 \cdot T_N$
A photo of a dog.



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I_1
$I_1 \cdot T_1$ $I_1 \cdot T_2$ $I_1 \cdot T_3$ ... $I_1 \cdot T_N$
A photo of a dog.
Contrastive Loss
(cosine similarity)
L^2 normalize
L^2 normalize
Contrastive Loss
(neg. Lorentzian distance)
+ Entailment Loss
expm_O
expm_O

2. Preliminaries

We briefly review Riemannian manifolds (Section 2.1) and essential concepts of hyperbolic geometry (Section 2.2). For a more thorough treatment of the topic, we refer the reader to textbooks by Ratcliffe (2006) and Lee (2019).

2.2. Lorentz model of hyperbolic geometry

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} = \langle \mathbf{x}_{space}, \mathbf{y}_{space} \rangle - x_{time}\, y_{time}$$

The induced *Lorentzian norm* is $\|\mathbf{x}\|_{\mathcal{L}} = \sqrt{|\langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}}|}$. The Lorentz model possessing a constant curvature $-c$ is defined as a following set of vectors:

$$\mathcal{L}^n = \{\mathbf{x} \in \mathbb{R}^{n+1} : \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -{}^1/c\ ,\ c > 0\} \qquad (2)$$

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$$d_{\mathcal{L}}(\mathbf{x}, \mathbf{y}) = \sqrt{{}^1/c} \cdot \cosh^{-1}(-c\, \langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}}) \qquad (4)$$

FRSE FRS FKC

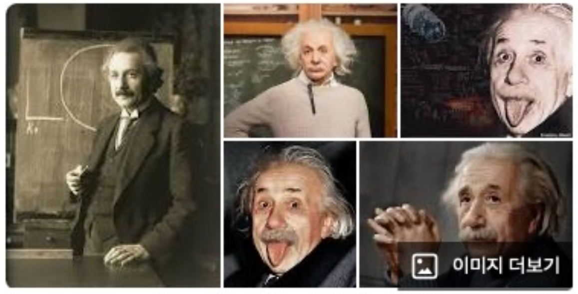
본명	제임스 클러크 맥스웰 James Clerk Maxwell
국적	영국
출생	1831년 6월 13일 그레이트브리튼 아일랜드 연합왕국 에든버러
사망	1879년 11월 5일 (향년 48세) 그레이트브리튼 아일랜드 연합왕국 케임브리지
직업	학자, 교수
분야	물리학, 수학

헤르만 민코프스키
Hermann Minkowski

출생	1864년 6월 22일 폴란드 왕국 수발키 알렉소타스

알베르트 아인슈타인

이론 물리학자



알베르트 아인슈타인은 독일 태생의 이론물리학자로서 역사상 가장 위대한 물리학자 중의 한 명으로 널리 알려져 있다. 상대성 이론을 개발한 것으로 유명하지만 양자역학 이론의 발전에도 중요한 공헌을 했다. 상대성 이론은 양자역학과 함께 현대 물리학의 두 기둥이다.
위키백과

출생: 1879년 3월 14일, 독일 울름

사망 정보: 1955년 4월 18일, 미국 뉴저지 프린스턴

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \varepsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

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알베르트 아인슈타인

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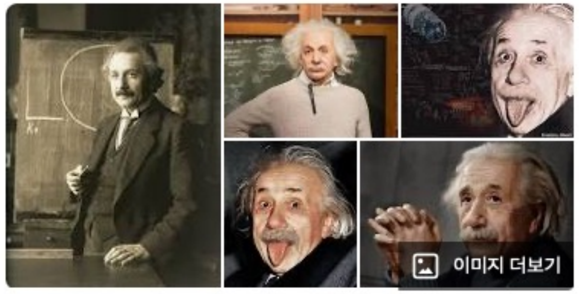

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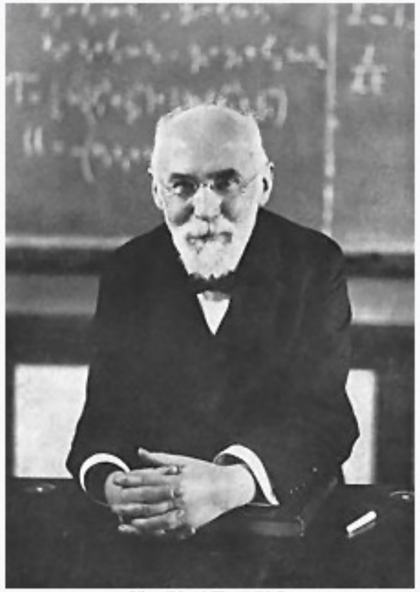
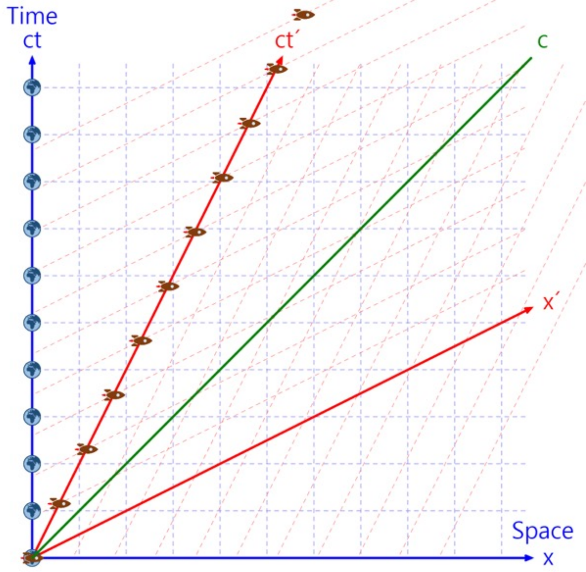
헨드릭 안톤 로런츠



헨드릭 안톤 로런츠

출생	1853년 7월 18일 네덜란드 아른험
사망	1928년 2월 4일 네덜란드 하를럼
국적	네덜란드
출신 학교	레이던 대학교
주요 업적	전자기 방사에 관한 이론 로런츠 힘 로런츠 변환 로런츠 인자 헤비사이드-로런츠 단위계
수상	노벨 물리학상 (1902)
분야	물리학
소속	레이던 대학교

$$x' = \gamma(x - ut)$$
$$y' = y$$
$$z' = z$$
$$t' = \gamma\left(t - \frac{ux}{c^2}\right)$$



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Exponential and logarithmic maps. The *exponential map* provides a way to map vectors from tangent spaces onto the manifold. For a point $\mathbf{z}$ on the hyperboloid, it is defined as $\operatorname{expm}_{\mathbf{z}} : \mathcal{T}_{\mathbf{z}}\mathcal{L}^n \to \mathcal{L}^n$ with the expression:

$$\mathbf{x} = \operatorname{expm}_{\mathbf{z}}(\mathbf{v}) = \cosh(\sqrt{c}\, \|\mathbf{v}\|_{\mathcal{L}})\, \mathbf{z} + \frac{\sinh(\sqrt{c}\, \|\mathbf{v}\|_{\mathcal{L}})}{\sqrt{c}\, \|\mathbf{v}\|_{\mathcal{L}}}\, \mathbf{v} \qquad (7)$$

Intuitively the exponential map shows how $\mathcal{T}_x\mathcal{L}^n$ *folds* on the manifold. Its inverse is the *logarithmic map* ($\operatorname{logm}_{\mathbf{z}} : \mathcal{L}^n \to \mathcal{T}_{\mathbf{z}}\mathcal{L}^n$), that maps $\mathbf{x}$ from the hyperboloid back to $\mathbf{v}$ in the tangent space:

$$\mathbf{v} = \operatorname{logm}_{\mathbf{z}}(\mathbf{x}) = \frac{\cosh^{-1}(-c\, \langle \mathbf{z}, \mathbf{x} \rangle_{\mathcal{L}})}{\sqrt{(c\, \langle \mathbf{z}, \mathbf{x} \rangle_{\mathcal{L}})^2 - 1}}\, \operatorname{proj}_{\mathbf{z}}(\mathbf{x}) \qquad (8)$$

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$$\mathcal{L}_{cont}$$

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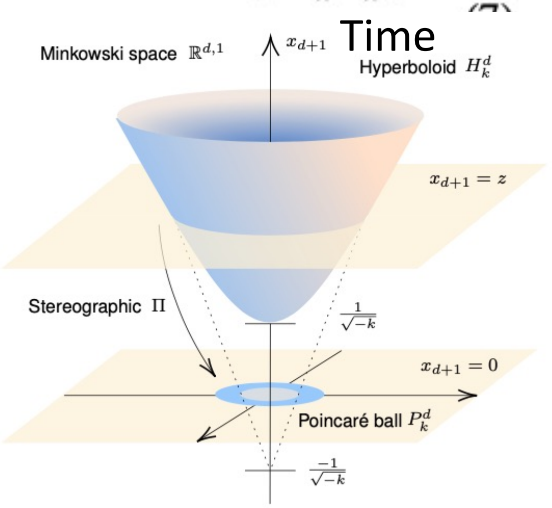
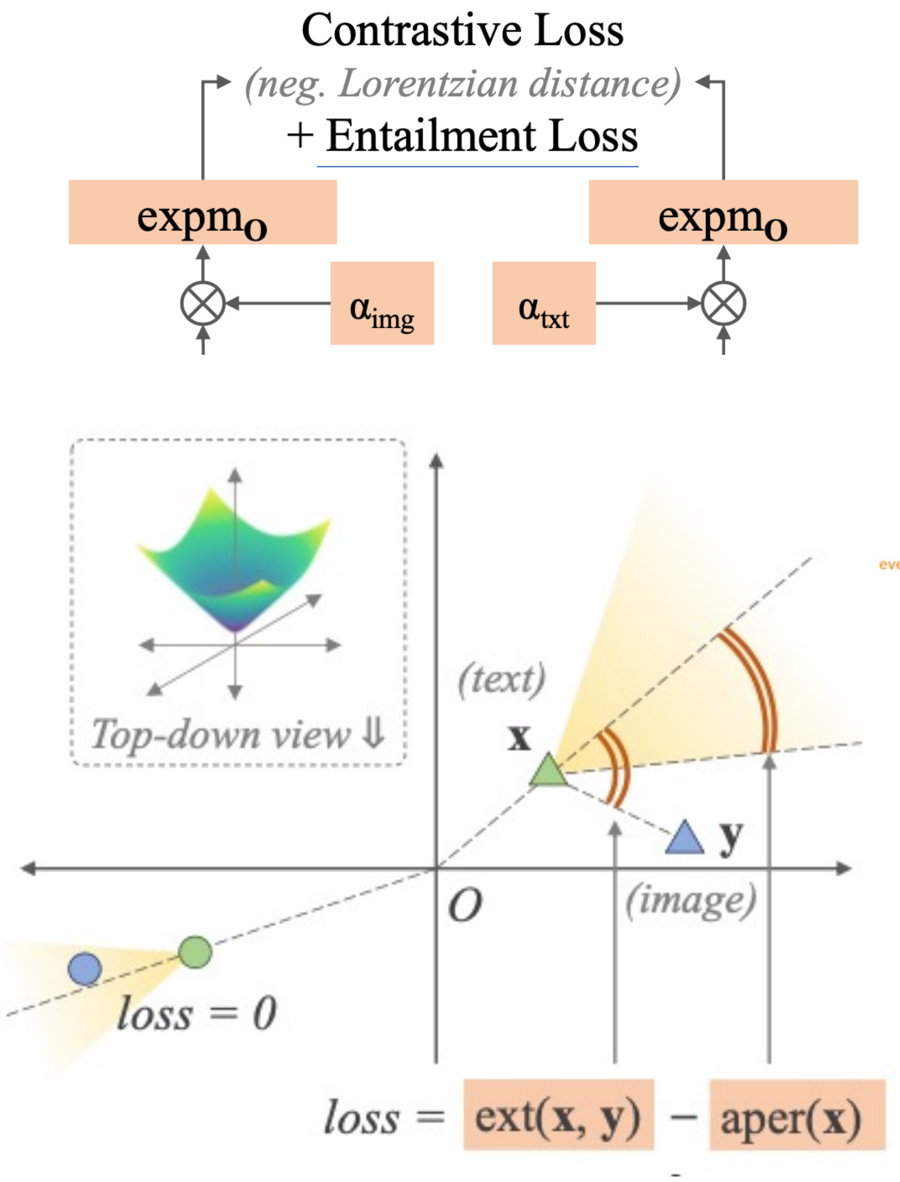
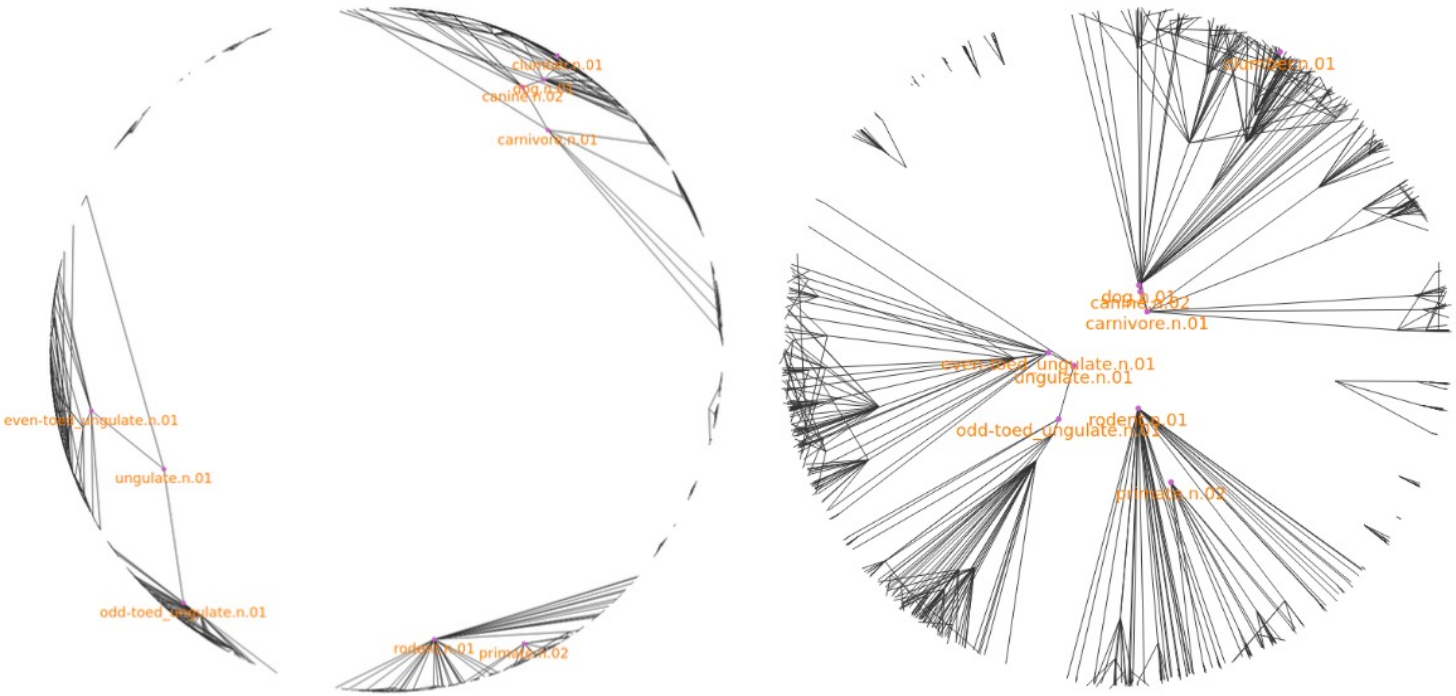
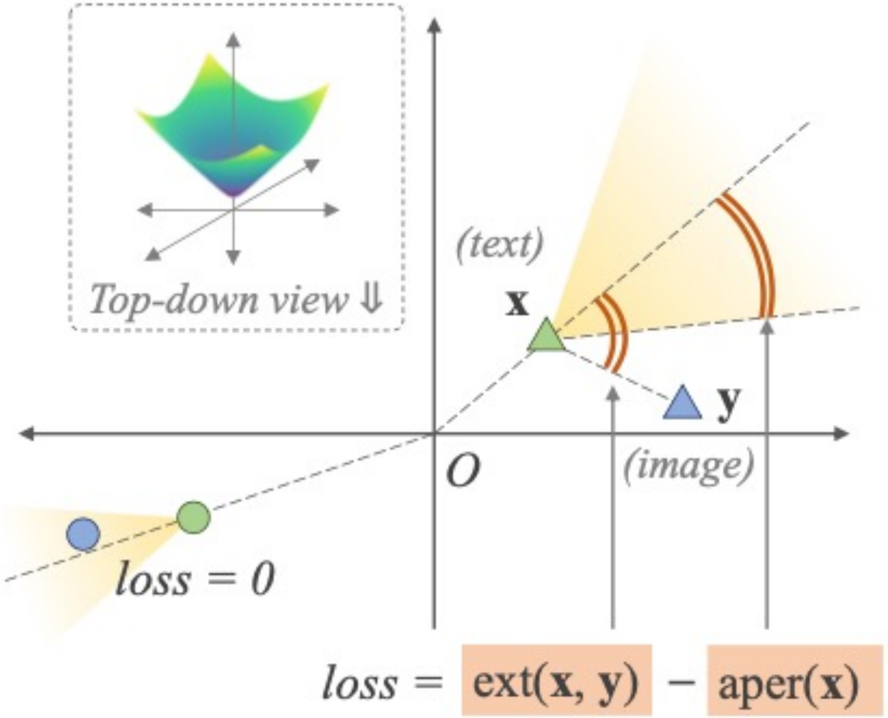
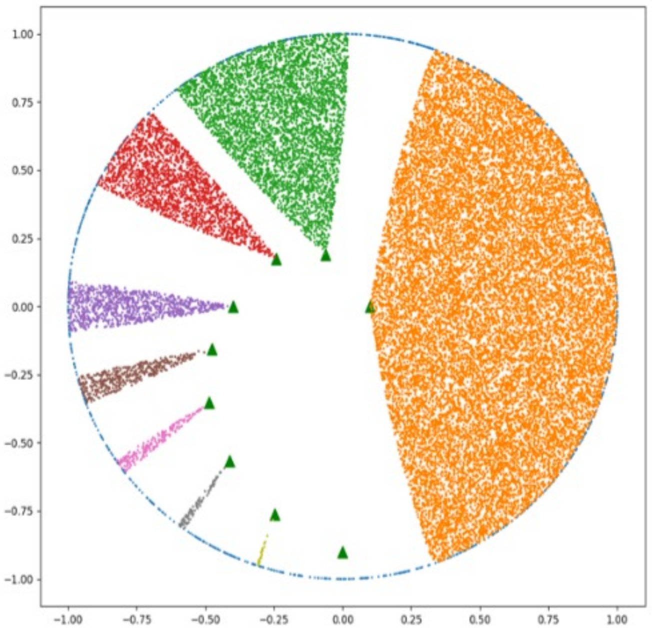






Hyperbolic Embedding VS Hyperbolic Entailment Cones 를 사용한 Embedding





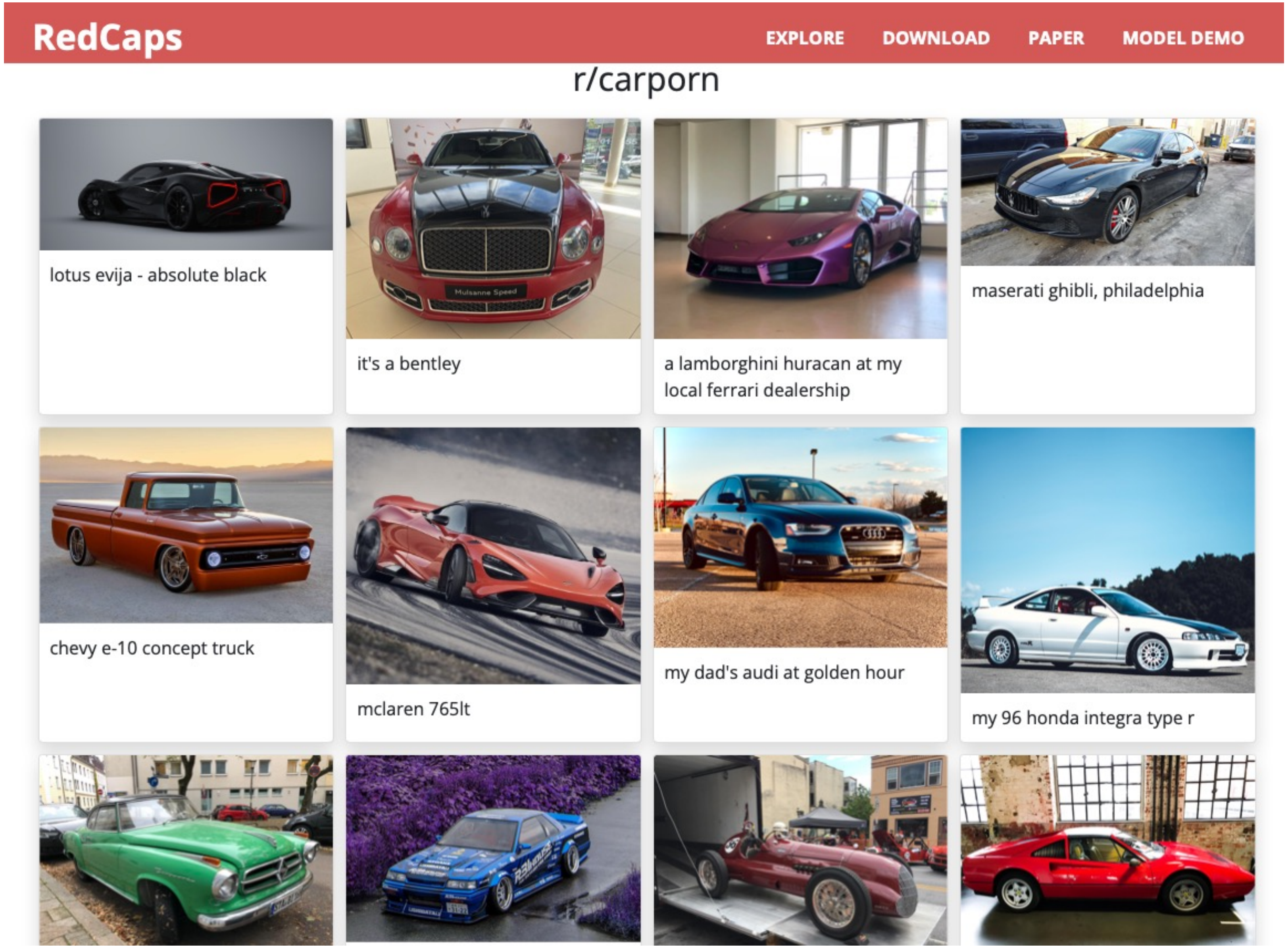
the half-aperture

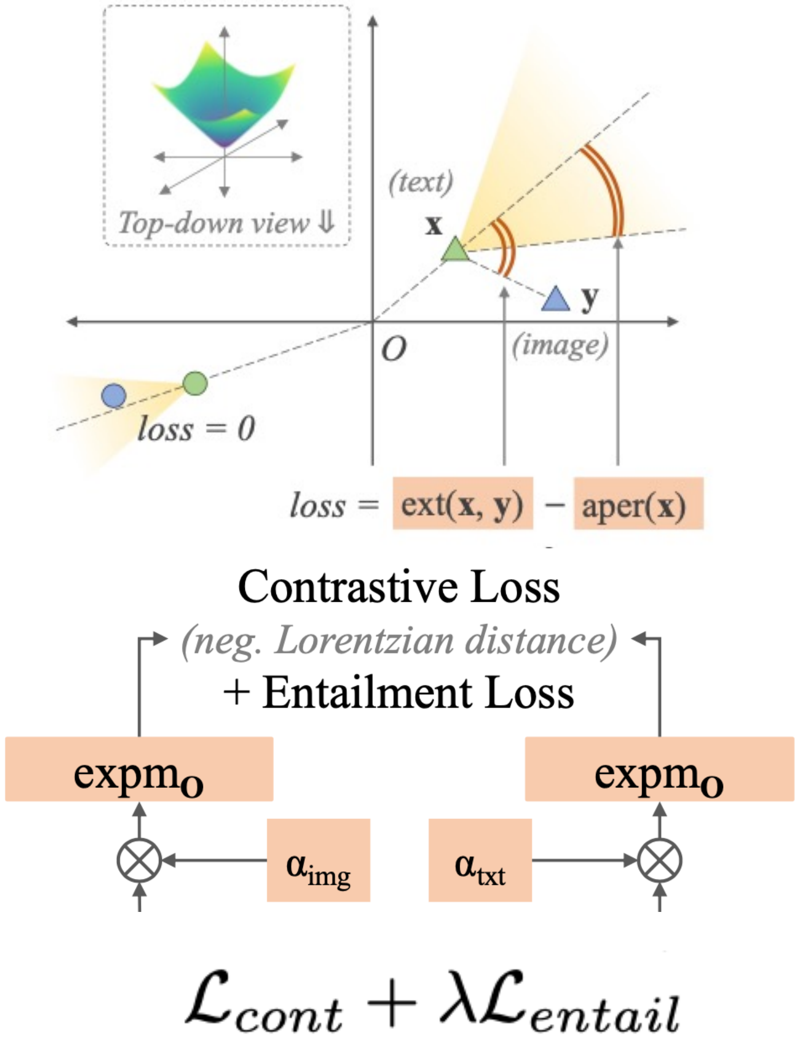
$$\text{aper}(\mathbf{x}) = \sin^{-1}\left(\frac{2K}{\sqrt{c}\,\|\mathbf{x}_{space}\|}\right)$$

$$\text{ext}(\mathbf{x},\mathbf{y}) = \cos^{-1}\left(\frac{y_{time} + x_{time}\, c\,\langle \mathbf{x},\mathbf{y}\rangle_{\mathcal{L}}}{\|\mathbf{x}_{space}\|\sqrt{\left(c\,\langle \mathbf{x},\mathbf{y}\rangle_{\mathcal{L}}\right)^2 - 1}}\right)$$

$$\mathcal{L}_{entail}(\mathbf{x},\mathbf{y}) = \max(0,\ \text{ext}(\mathbf{x},\mathbf{y}) - \text{aper}(\mathbf{x}))$$

$$\mathcal{L}_{cont} + \lambda\mathcal{L}_{entail}$$


RedCaps
EXPLORE
DOWNLOAD
PAPER
MODEL DEMO
r/carporn
lotus evija - absolute black
it's a bentley
a lamborghini huracan at my local ferrari dealership
maserati ghibli, philadelphia
chevy e-10 concept truck
mclaren 765lt
my dad's audi at golden hour
my 96 honda integra type r



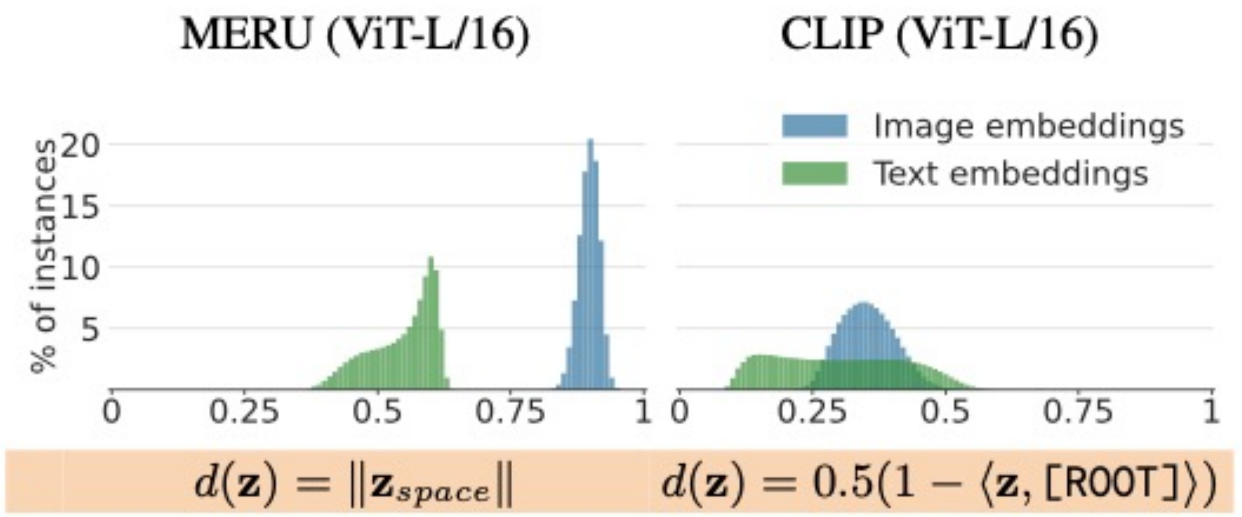


Figure 4. **Distribution of embedding distances from `[ROOT]`:** We embed all 12M training images and text using trained MERU and CLIP. Note that precise distance is not necessary for this analysis, so we compute simple monotonic transformations of distances, $d(\mathbf{z})$. MERU embeds text closer to `[ROOT]` than images.

Table 2. **Zero-shot image classification.** We train MERU and CLIP models with varying parameter counts and transfer them *zero-shot* to 20 image classification datasets. Best performance in every column is highlighted in green. Hyperbolic representations from MERU match or outperform CLIP on 13 out of the first 16 datasets. On the last four datasets (gray columns), both MERU and CLIP have *near-random* performance, as concepts in these datasets are not adequately covered in the training data.

		ImageNet	Food-101	CIFAR-10	CIFAR-100	CUB	SUN397	Cars	Aircraft	DTD	Pets	Caltech-101	Flowers	STL-10	EuroSAT	RESISC45	Country211	MNIST	CLEVR	PCAM	SST2
ViT S/16	CLIP	34.3	74.5	**60.1**	24.4	**33.8**	27.5	**11.3**	**1.4**	15.0	**73.7**	63.9	47.0	88.2	18.6	31.4	**5.2**	10.0	19.4	50.2	50.1
	MERU	**34.4**	**75.6**	52.0	**24.7**	33.7	**28.0**	11.1	1.3	**16.2**	72.3	**64.1**	**49.2**	**91.1**	**30.4**	**32.0**	4.8	7.5	14.5	51.0	50.0
ViT B/16	CLIP	**37.9**	**78.9**	65.5	**33.4**	33.3	29.8	**14.4**	1.4	17.0	77.9	**68.5**	50.9	92.2	25.6	31.0	**5.8**	10.4	14.3	54.1	51.5
	MERU	37.5	78.8	**67.7**	32.7	**34.8**	**30.9**	14.0	**1.7**	**17.2**	**79.3**	**68.5**	**52.1**	**92.5**	**30.2**	**34.5**	5.6	13.0	13.5	49.8	49.9
ViT L/16	CLIP	38.4	80.3	**72.0**	**36.4**	36.3	32.0	**18.0**	1.1	16.5	78.8	**68.3**	48.6	**93.7**	26.7	35.4	6.1	14.8	13.6	51.2	51.1
	MERU	**38.8**	**80.6**	68.7	35.5	**37.2**	**33.0**	16.6	**2.2**	**17.2**	**80.0**	67.5	**52.1**	**93.7**	**28.1**	**36.5**	**6.2**	11.8	13.1	52.7	49.3

Table 1. **Zero-shot image and text retrieval.** Best performance in every column is highlighted in green. MERU performs better than CLIP for both datasets and across all model sizes.

		text → image				*image → text*			
		COCO		Flickr		COCO		Flickr	
		R5	R10	R5	R10	R5	R10	R5	R10
ViT S/16	CLIP	29.9	40.1	35.3	46.1	37.5	48.1	42.1	54.7
	MERU	**30.5**	**40.9**	**37.1**	**47.4**	**39.0**	**50.5**	**43.5**	**55.2**
ViT B/16	CLIP	32.9	43.3	40.3	51.0	41.4	52.7	**50.2**	**60.2**
	MERU	**33.2**	**44.0**	**41.1**	**51.6**	**41.8**	**52.9**	48.1	58.9
ViT L/16	CLIP	31.7	42.2	39.0	49.3	40.6	51.3	47.8	58.5
	MERU	**32.6**	**43.0**	**39.6**	**50.3**	**41.9**	**53.3**	**50.3**	**60.6**

		Embedding width				
		512	256	128	96	64
COCO *text→image*	CLIP	31.7	31.8	31.4	29.6	25.7
	MERU	**32.6**	**32.7**	**32.7**	**31.0**	**26.5**
COCO *image→text*	CLIP	40.6	41.0	40.4	37.9	33.3
	MERU	**41.9**	**42.5**	**42.6**	**40.5**	**34.2**
ImageNet	CLIP	38.4	38.3	37.9	35.2	30.2
	MERU	**38.8**	**38.8**	**38.8**	**37.3**	**32.3**

Table 3. **MERU and CLIP with different embedding widths.** We report *zero-shot* COCO recall@5 and ImageNet top-1 accuracy. MERU outperforms CLIP at lower embedding widths.

MERU	CLIP
seagull	*seagull*
bird	*bird*
air	↓
coast	↓
day	↓
[ROOT]	[ROOT]

MERU	CLIP
a bengal cat sitting beside wheatgrass on a white surface	*a bengal cat sitting beside wheatgrass on a white surface*
bengal	↓
cat	↓
domestic	↓
[ROOT]	[ROOT]

MERU	CLIP
white horse	*white horse*
equine	↓
equestrian	↓
beauty	↓
female	↓
fluffy	↓
[ROOT]	[ROOT]

MERU	CLIP
photography of rainbow during cloudy sky	*phenomenon*
rainbow	↓
phenomenon	↓
rural	↓
[ROOT]	[ROOT]

MERU	CLIP
retro photo camera on table	↓
fujinomiya	↓
vintage	↓
style	↓
[ROOT]	[ROOT]

MERU	CLIP
avocado toast	*avocado toast*
healthy breakfast	*delicious*
delicious	↓
homemade	↓
fresh	↓
[ROOT]	[ROOT]

MERU	CLIP
brooklyn bridge	*photo of brooklyn bridge, new york*
new york city	*new york city*
city	*new york*
outdoors	↓
day	↓
[ROOT]	[ROOT]

MERU	CLIP
taj mahal	*taj mahal through an arch*
monument	*travel*
architecture	*inspiration*
travel	↓
day	↓
[ROOT]	[ROOT]

MERU	CLIP
sydney opera house	*sydney opera house*
opera house	*opera house*
holiday	*gift*
day	*beauty*
[ROOT]	[ROOT]

Authors choose vector dimensionality n to be 50 and J to be 5. $\mathbf{x}_0 = \mathbf{0}, \sigma_0 = 1, \sigma_j = \frac{1}{5}$. Branching factor is set to $b = 2$ and maximum depth is $d = 6$.

Results of using Poincaré VAE on this dataset can be seen in fig. 3.7.

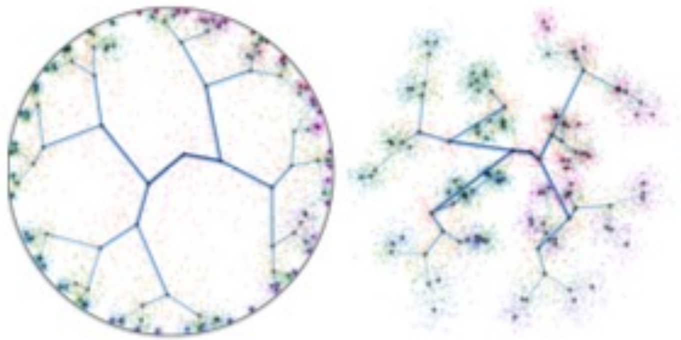


Fig. 3.7.: Embedding of branching diffusion process using Poincaré VAE (left) and standard VAE (right); reproduced from Mathieu et al. (2019).

The suggested way of creating hierarchical dataset will come extremely useful in the later chapters since we have taken it as an inspiration for creating tree-like graph dataset.